
Dr. Tanima Mondal

Postdoctoral Research Associate - I, Aryabhata Research Institute of Observational Sciences (ARIES), Nainital, India

Ph.D. (Multi-messenger Astrophysics), Department of Physics, IIT Kharagpur, Kharagpur, West Bengal, India

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Research Expertise

Multi-Messenger Astrophysics, High energy Astrophysics, Gamma-ray Bursts, Astroparticle physics, CTAO Detectability, GW-neutrino follow-up from transients, Machine/Deep Learning in event reconstruction and classification.

Awarded Doctoral degree on 17th March, 2026.

Research Works

Thesis Title — Multi-messenger detection prospect of Gamma-ray Burst Afterglow at very high energy regime 2020(January) – 2025(December)

Dept. of Physics, IIT Kharagpur

- Developed long and short GRB afterglow models at the VHE regime, analysed and validated on real observational data (radio, optical, X-ray, GeV-TeV).
- Predicted detectability of short GRB–GW events in the VHE regime using LIGO and CTAO Northern array sensitivity.
- Developed photo-hadronic neutrino models to assess neutrino emission prospects of IceCube Gen2/GRAND 200k detection probability.
- Built structured jet models in wind-driven medium to analyse spectral evolution and detectability in VHE gamma-ray bands.
- Developing a deep-learning Convolutional Neural network (CNN ResNet-18) based analysis framework for electron neutrino event classification and reconstruction of kinematic variables for the Hyper-K neutrino experiment.

Photometric Observations and Stellar Cluster Analysis

2018 – 2019

Fr. Eugene Lafont Observatory, St. Xavier's College Kolkata

- Observed and analyzed open clusters NGC2301, M35, and M45.
 - Performed differential photometry and data reduction on FITS images.
 - Contributed to the setup and commissioning of the Eugene Lafont Observatory, Kolkata, India.
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Technical Skills

Programming and Software: Python, C++, MATLAB, ROOT, GEANT4, WcSim, DAOPHOT, WSClean, CASA, XSPEC, ThreeML, Gammapy, CTLearn, JetSeT, PyTorch.

Scientific Expertise: Multi-messenger astrophysics (GRBs, GW, BNS merger, neutrinos), relativistic jet and afterglow modeling, synchrotron and SSC radiation processes, high-energy data analysis (radio, optical, XRT, Fermi GBM/LAT), statistical inference and likelihood analysis.

Computational and ML Methods: Numerical ODE/PDE solvers, simulation-based modeling, convolutional neural networks (ResNet-18), machine learning algorithm for event classification and reconstruction frameworks.

Work Experience

Research Intern

May 2022 – Aug 2022

TRIUMF, Canada

- Optimized the fitQun reconstruction algorithm for IWCD detector's updated event geometry.
- Applied ML methods to distinguish e^- , μ^- , π^0 and γ -like events with improved resolution.

Visiting Researcher

July 2018 – Nov 2019

Dept. Of Physics, St. Xavier's College Kolkata

- I planned and taught undergrad and grad students on Star Cluster Photometry, details of Hertzsprung Russel diagram and mentored related experimental works.
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Education

Ph.D. in Physics

Jan 2020 – Dec 2025

Indian Institute of Technology (IIT) Kharagpur

• Coursework Completed:

- **PH60044: High Energy Physics** — Taught by Prof. Utpal Sarkar
- **TS70006: Quantum Mechanics and Quantum Computing** — Taught by Prof. Bibhas Adhikari and Prof. Sonjoy Majumder
- **PH60046: Astrophysics and Cosmology** — Taught by Prof. Somnath Bharadwaj
- **TS62001: Simulation in Collider Physics and Cosmology** — Taught by Prof. Somnath Bharadwaj and Prof. Debanjan Bose
- **TS70007: Advanced Mathematical Techniques** — Taught by Prof. S. P. Khastgir, Prof. R. Kannan, and Prof. H. S. Mahato
- **HS63002: English for Technical Writing** — Taught by Prof. B. Lahiri and Prof. D. Piplai

M.Sc. in Physics

2016 – 2018

St. Xavier's College, University of Calcutta

- Master's project: Photometric observation and analysis of open clusters (NGC2301, M35, M45).
 - Gained expertise in observational astronomy, FITS data analysis.
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Conference Presentations (Oral)

- ICRC 2025: Multi-Messenger Detection and Neutrino Emission Prospects of gamma ray burst Afterglows from Binary Neutron Star Mergers
 - ICRC 2023: Modeling of Gamma Ray Burst (GRB) Afterglow at Very High Energy (VHE) regime
 - Hyper-K WCTE 2022: ML-based particle identification (PID) Studies for event sample at Intermediate Water Cherenkov Detector (IWCD)
 - Indo-Japan HyperK 2022 workshop: ν_e Sample Analysis with Deep Learning for Hyperk Experiment
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Poster Presentations

- ICHEP 2024 (**Best Poster**): Likelihood and Deep Learning Analysis of the electron neutrino event sample at Intermediate Water Cherenkov Detector (IWCD) of the Hyper-Kamiokande experiment
 - ASI 2024: Modelling very high energy GRB afterglow emission
 - DAE-BRNS 2022: Modelling very high energy GRB afterglow emission
 - Hyper-K Collab. 2022: ML Techniques for HyperK Event Reconstruction
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Teaching Assistantships

- **External:**
 - HIJLI COLLEGE (Sem-VII October 2024): I taught the Research Methodology (Scientific Plotting using Python) in Hijli College.
 - HIJLI COLLEGE (Sem-VI February 2024): I taught the Research Methodology (Fundamentals Of Machine Learning) in Hijli College following their structured syllabus.
 - HIJLI COLLEGE (Sem-VII August 2023): I taught the Elementary ideas of Machine Learning under Research Methodology in Hijli College.
 - HIJLI COLLEGE (Sem-VI February 2023): I taught the CC-13: Electromagnetic Theory in Hijli College following their structured syllabus.

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- HIJLI COLLEGE (Sem-V August 2022): I taught the Modern Physics course (Particle physics and Detector for Nuclear Radiations) in Hijli College following their structured syllabus.
 - HIJLI COLLEGE (Sem-IV March 2022): I taught the Nuclear Physics course on semester-4 in Hijli College following their structured syllabus.

- **Internal:**

- Internal TA duty at Dept. Of Physics, IIT Kharagpur —
- Computational Lab classes for M.Sc (Course name - (PH39209))
- Electromagnetism and Optics Lab(PH39205)
- Spectroscopy and Electromagnetism Lab (PH29203/PH29207)

Conference, Workshops and Schools Attended

- ICRC 2025
- ICHEP 2024
- ASI 2024
- Understanding Universe Through Neutrinos, ICTS, 2024
- ICRC 2023
- ICTS - School on Statistical Methods and Machine Learning in High Energy Physics, 2023
- DAE-BRNS Symposium 2022
- Indo-Japan HyperK Workshop 2022
- HyperK-WCTE 2022 Meeting
- TRISEP 2022, TRIUMF, Canada
- Astronomy Winter School, NTHU/NCTS 2021
- PHYSTAT Systematics Workshop 2021
- ASI 2021
- IUAC School on Detector Simulation using GEANT4, 2021
- GEANT4 Detector Simulation School, IUAC 2021
- DAE-BRNS Symposium 2020
- ICTS, Gravitational Wave Astrophysics (ONLINE), 2020

Awards and Recognition

- Best Poster Award, ICHEP 2024
- Prime Minister's Research Fellowship, 2021
- Mitacs Globalink Research Award
- GATE Qualified, 2019
- IIT(ISM) M.Sc. Entrance Qualified, 2016
- Merit-cum-Means Scholarships (Secondary and HS)

List of publications

1. K. Abe, P. Adrich, [...], **T. Mondal** et al, *Supernova Model Discrimination with Hyper-Kamiokande*, [The Astrophysical Journal](#) **916**, 15 (2021)^{†,★}.
2. Chakraborty, S., **T. Mondal**, A. Debnath, and S. Roychowdhury, *A multiresolution approach to enhance small telescope data under non-ideal conditions*, [Journal of Astrophysics and Astronomy](#) **43**(1), 22 (2022).
3. **T. Mondal**, S. Pramanick, L. Resmi, and D. Bose, *Probing gamma-ray burst afterglows with the Cherenkov Telescope Array*, [Monthly Notices of the Royal Astronomical Society](#) **522**, 5690–5700 (2023)^{†,★}.
4. **T. Mondal**, L. Resmi, and D. Bose, *Modeling of Gamma Ray Burst (GRB) Afterglow at Very High Energy (VHE) regime*, [PoS ICRC2023](#), 600 (2023)[†].
5. **T. Mondal**, N. W. Prouse, P. de Perio, M. Hartz, and D. Bose, *Likelihood and Deep Learning Analysis of the electron neutrino event sample at Intermediate Water Cherenkov Detector (IWCD) of the Hyper-Kamiokande experiment*, [PoS ICHEP2024](#), 232 (2024)^{†,★}.
6. **T. Mondal**, S. Chakraborty, L. Resmi, and D. Bose, *Follow-up of Neutron Star Mergers with CTA and Prospects for Joint Detection with Gravitational-wave Detectors*, [The Astrophysical Journal](#) **983**, 35 (2025)^{†,★}.
7. **T. Mondal**, S. Chakraborty, J. Joshi, L. Resmi and D. Bose, *Multi-Messenger Detection and Neutrino Emission Prospects of gamma ray burst Afterglows from Binary Neutron Star Mergers*, [PoS ICRC2025](#), 952 (2025)[†].
8. **T. Mondal**, S. Razzaque, Jagdish C. Joshi, and D. Bose, *Multi Messenger Study of GRB 221009A with VHE Gamma-ray and Neutrino Afterglow from a Gaussian Structured Jet*, [https://doi.org/10.1016/j.jheap](https://doi.org/10.1016/j.jheap.2026.01.001) (2026)^{†,★}.
9. **T. Mondal**, S. Chakraborty, L. Resmi, and D. Bose, *Probing TeV Afterglow Emission of GRB 221009A with Gaussian Structured jet in Wind-driven medium*, [arXiv:2512.04893 \[astro-ph.HE\]](#) (2025)[†].
10. K. Abe, M. T Afif, [...], **T. Mondal** et al. *Sensitivity of the Hyper-Kamiokande experiment to neutrino oscillation parameters using accelerator neutrinos*, [The European Physical Journal C](#) **86**, 176 (2026).

[†] Paper used for the thesis ★ Main papers
